IMPULSIVE ONLINE PURCHASE

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Background

Business Question: **"Can We Predict Which Online Shoppers Are Likely to Make Impulsive Purchases?"**

Motivation

- Impulsive Buying Impacts:
  - Business revenue
  - Consumer behavior
  - Marketing strategy

Significance

- Online shopping keeps growing ~ 18% of 2025 US GDP
- Companies care about conversion rates
- Helps understand digital consumer psychology

Shopping Behavior Data

Cleaning the Data:

- converted purchase frequency into numeric values to create meaningful quantitative feature
- removed irrelevant and non-predictive columns
- one hot-encoded categorical variables
  - multicollinearity, clean binary columns, dataset fully numeric & model ready

Daily	30
Weekly	7
Monthly	1
Quarterly	0.25

Creating the Y Variable (Impulsive_Purchase):

- 0 = Not Impulsive, 1 = Impulsive

Shopper was labeled impulsive if...

- shopped monthly or more, **OR** buy more than avg. # of prev. purchases
- **AND** if they used a discount or promo code

⇒ Impulsive shopping behavior : frequent purchases + incentive-triggered buying

- **detect spending patterns**
- **identify discount driven shoppers**
- **measure purchase frequency**
- **recognize behavioral triggers**

The definition of y variable can be adjusted according to the business model

Logistic Regression (feature selection)

These traits consistently separate
impulsive vs. non-impulsive shoppers
(variables across all models)

- Purchase Amount
- Review Rating
- Previous Purchases
- Gender
- Location (Alaska, Arizona, Colorado, Connecticut)
- Season
- Subscription
- Shipping Type
- Payment Method

Logistic Regression (full model)
- Total of 72 features

Logistic Regression (forward selection)
- 35 features are selected

Logistic Regression (backward selection)
- 44 features are selected

What Did These Features Teach Us about Impulsive Online Shoppers?

- subscription status & convenience choices are significant
- seasonal & demographic differences influence impulsive buying tendencies
- past behavior (frequency + amount) was the most consistent factor

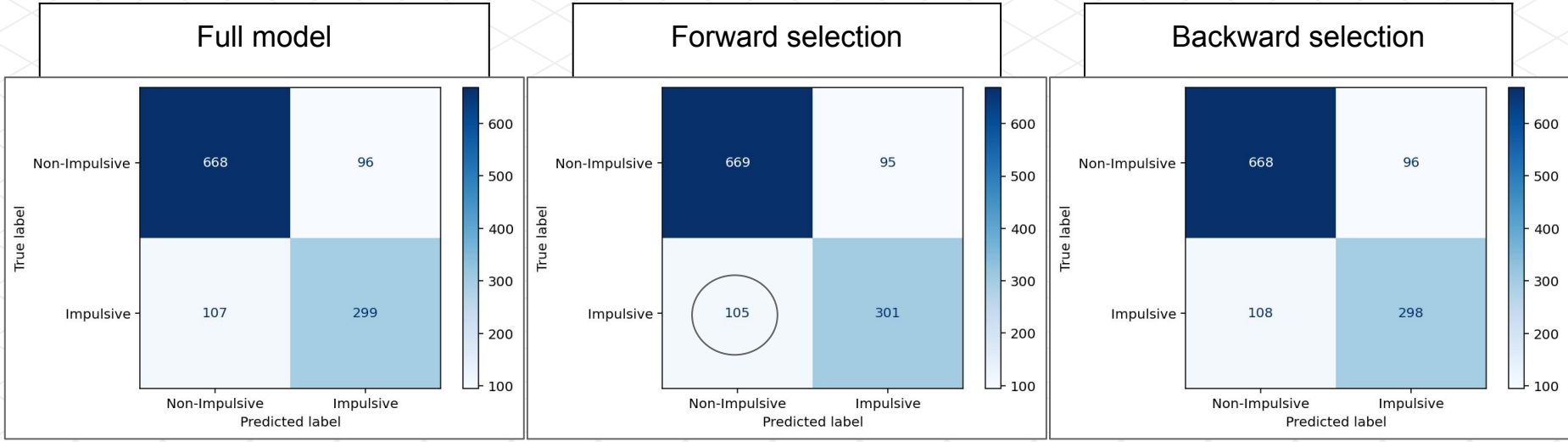
Logistic Regression (scores)

- F1 training < F1 testing → not overfitting
- Forward selection has the highest scores among all the models

Metric	Full model	Forward	Backward
F1 (training)	0.7239	0.7258	0.7244
F1 (testing)	0.7466	0.7506	0.7450
Precision	0.7570	0.7601	0.7563
Accuracy	0.8265	0.8291	0.8256
Recall	0.7365	0.7414	0.7340

Logistic Regression (confusion matrix)

- False Negative (FN): Saying a purchase is NOT impulsive when it is - is most costly (missed opportunity)
- Forward selection has the least amount of False Negative results



Shopping Behavior Decision Tree

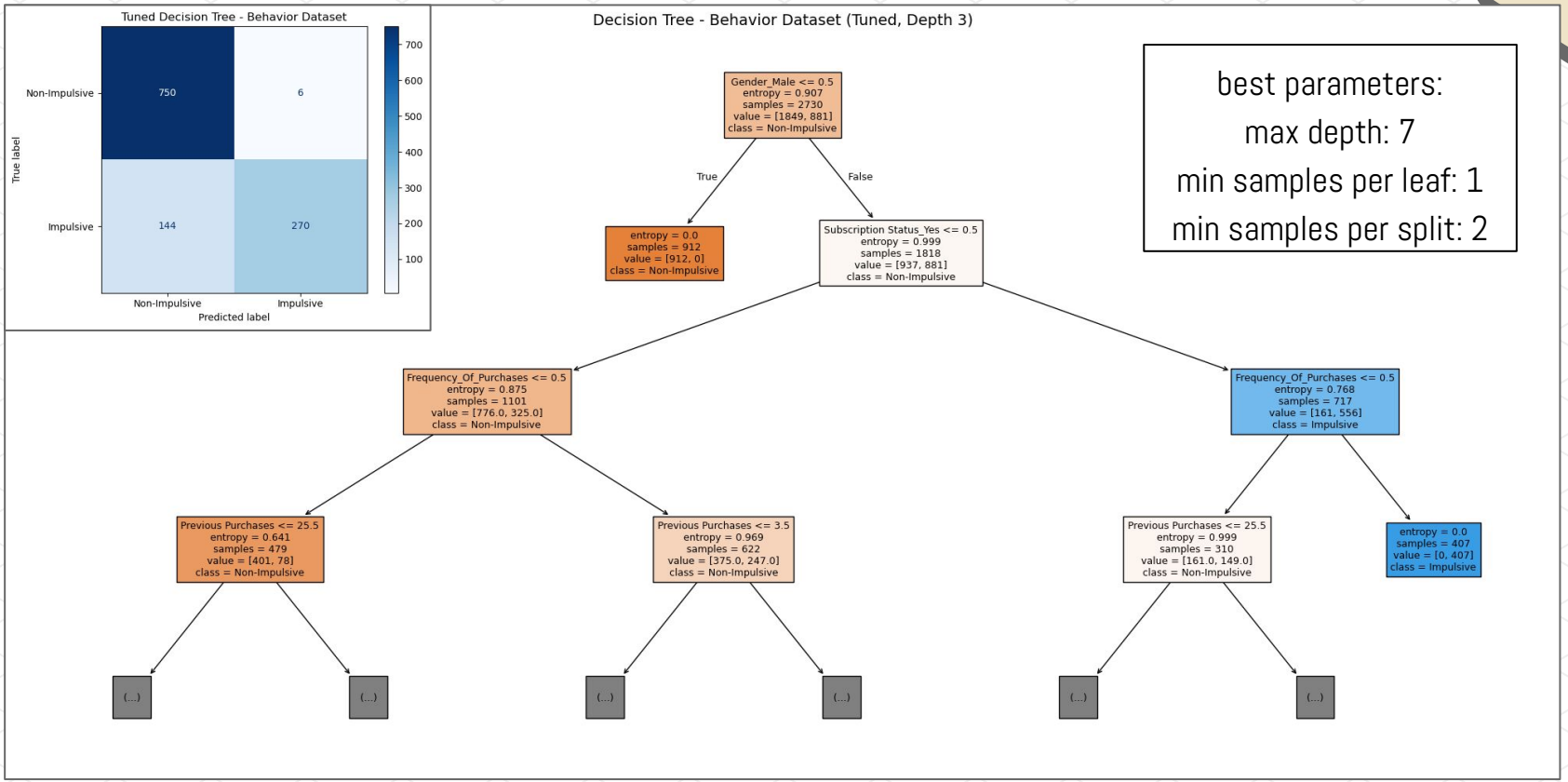
Metrics	Untuned	Tuned
Accuracy	0.85	0.87
Precision	0.8	0.98
Recall	0.78	0.65
Training F1	1	0.79
Testing F1	0.79	0.78

- training f1 score > testing f1 score → overfitting
- in the tuned model
  - overfitting was reduced
  - model accuracy & precision increased
- lower recall in tuned model

What Did The Performance Teach Us about Impulsive Online Shoppers?

- Some impulsive buyers behave consistently (habit-driven), others impulsively only situationally.
- Impulsive purchases follow a structured pattern; the model identified behavior rules.
- Impulsive shoppers aren't one-type: some are predictable habit-buyers while others require the right trigger.

Shopping Behavior Decision Tree



best parameters:

max depth: 7

min samples per leaf: 1

min samples per split: 2

Shopping Behavior Random Forest

Metrics	Untuned	Tuned
Accuracy	0.87	0.87
Precision	0.96	0.97
Recall	0.66	0.65
Training F1	1	1
Testing F1	0.78	0.78

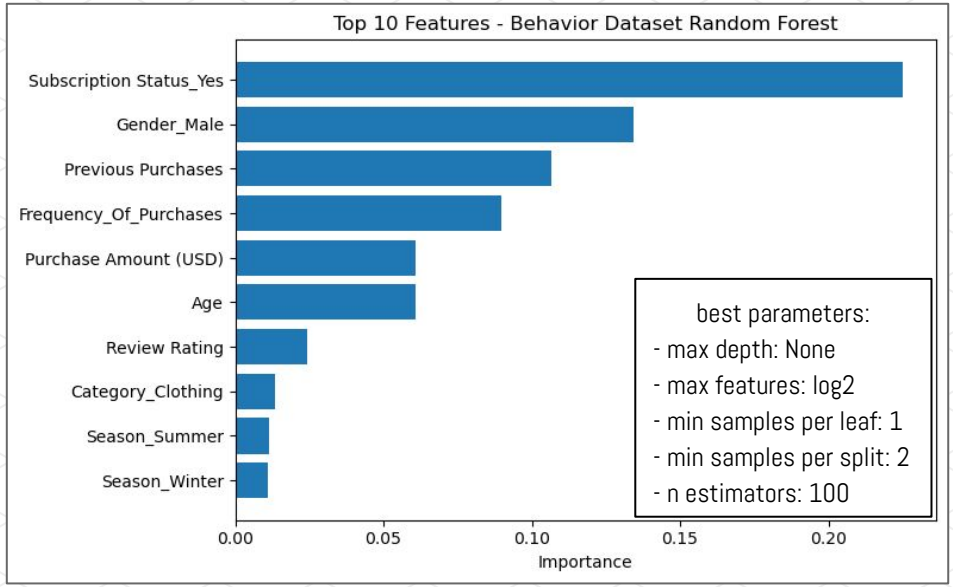
⇒ overfitting, not resolved by tuning
⇒ tuned model had slightly better accuracy and precision scores
⇒ low recall
⇒ not much better performing than decision tree

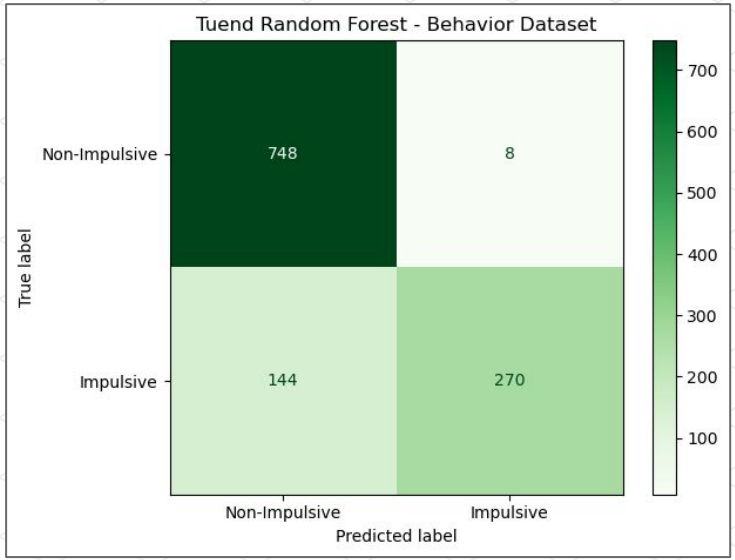
What Did The Performance Teach Us about Impulsive Online Shoppers?

- Impulsive shoppers share a strong behavioral signature detectable across many splits.
- Even with tuning, recall stayed low ⟶ impulsivity exists in more than one form.
- Suggests two shopper profiles: frequent predictable impulsives + occasional situational impulsives.

Shopping Behavior Random Forest

Repeat & Subscribed Customers are the Most Impulse-Prone Group





Comparing Models: What Do They Mean Together?

Metrics	Decision Tree (Tuned)	Random Forest (Tuned)
Accuracy	0.87	0.87
Precision	0.98	0.97
Recall	0.65	0.65
Training F1	0.79	1
Testing F1	0.78	0.78

⇒ overall the models performed pretty similarly

⇒ the tuned decision tree model is better fitted than the random forest model and has a slightly better precision score

Both models performed nearly identically: impulsive buying behavior is stable & predictable.

Two Impulsive Buying Styles: Impulsive buying is consistent enough to be predicted but diverse enough to require different targeting strategies.

KNN (scores)

Metrics	Before Tuning	After Tuning
F1(training)	0.80	0.80
F1 (testing)	0.64	0.69
Precision	0.71	0.82
Recall	0.59	0.59
Accuracy	0.78	0.82

- training F1 was strong, but testing F1 dropped
- High precision + lower recall = the model catches the "obvious impulsives" but misses subtle ones

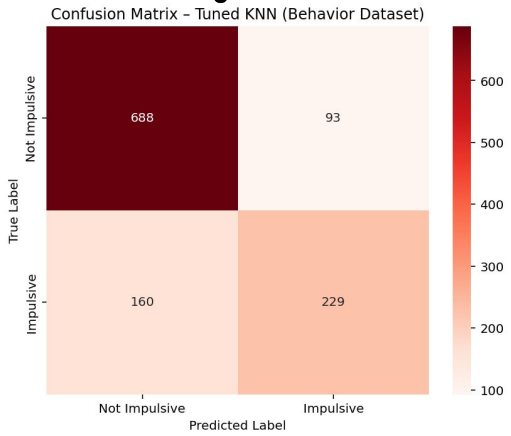
What Did The Performance Teach Us about Impulsive Online Shoppers?

Impulsive buying patterns exist but are not universal.

Supports Impulsive Buying Has Two Layers: habit-driven impulsivity (predictable) & trigger-based impulsivity (harder to detect).

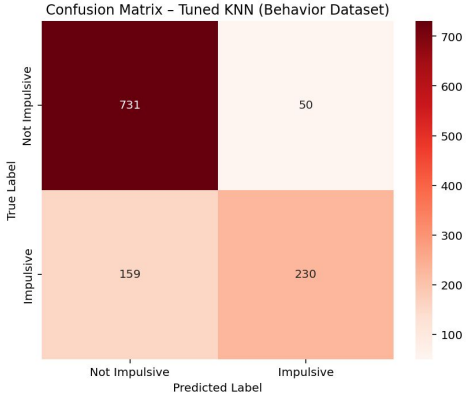
KNN - Confusion Matrix

Before Tuning



- KNN correctly identifies most non-impulsive shoppers: their behavior patterns are stable.
- More impulsive shoppers were misclassified: impulsivity varies and isn't always visible through past actions.

After Tuning ('b__n_neighbors': 19, 'b__p': 1)



Impulsive behavior is less uniform than non-impulsive behavior, and some buyers only appear impulsive at the moment of purchase.

Shopping Behavior - Conclusion

1. Impulsive buying is predictable: patterns exist & models confirmed them.
2. Two impulsive buyer types emerged: predictable habitual buyers & unpredictable trigger-based ones.

Metrics	Logistic Regression (forward)	Decision Tree	KNN
F1 (training)	0.7258	0.788	0.8
F1 (testing)	0.7506	0.783	0.69
Precision	0.7601	0.978	0.82
Accuracy	0.8291	0.872	0.82
Recall	0.7414	0.652	0.59

Two-Type Shopper Diagram:
Impulsive Shoppers

Predictable Impulsives
- frequent buyers
- promo-reactive
- subscription heavy
- easy to target

Situational Impulsives
- irregular buyers
- triggered by mood/event
- no clear history
- timing-dependent

3. **Impulsive buying is patterned & situational:** the predictable group can be targeted, while the spontaneous group can be influenced.

Mendeley Dataset BG

GOAL: Determine Psychological Impulsivity

Dataset Structure: 306 Respondents,

Y Variable
Impulsive Buying Behavior
(IBB1-IBB4 → mean score)

X Variables
promotion, social influence, happiness, self-control, normative evaluation, and demographics

- ☆ checked for missing values
- ☆ converted Likert-scale responses into numerical format
- ☆ combined impulsive-buying survey questions into a single target
- ☆ aggregated psychological features into constructs (x variables)

captures internal drivers of impulsive buying: motivations, emotions, social influence, and self-control.

- understand **the psychology behind impulsive decisions** , not just behavior patterns.

Logistic Regression (feature selection)

These traits consistently separate
impulsive vs. non-impulsive shoppers
(variables across all models)

- Monthly Income
- Self Control
- Social Influence
- Promotion

- Logistic Regression (full model)
  - Total of 12 features
- Logistic Regression (forward selection)
  - 4 features are selected
- Logistic Regression (backward selection)
  - 9 features are selected

What Did These Features Teach Us about Impulsive Online Shoppers?

- Promotion sensitivity is the strongest predictor of impulsivity
- Happiness & positive emotion increase spontaneous spending
- Self-control acts inversely:
  more control = less impulsive buying

Logistic Regression (scores)

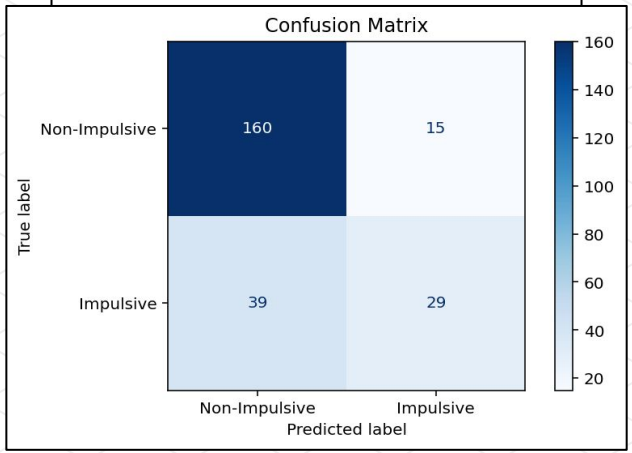
- F1 training > F1 testing → slightly overfitting
- Backward selection has the highest scores among all the models

Metrics	Full model	Forward	Backward
F1 (training)	0.5747	0.5440	0.5747
F1 (testing)	0.5179	0.5185	0.5487
Precision	0.6591	0.700	0.6889
Accuracy	0.7778	0.7860	0.7901
Recall	0.4265	0.4118	0.4559

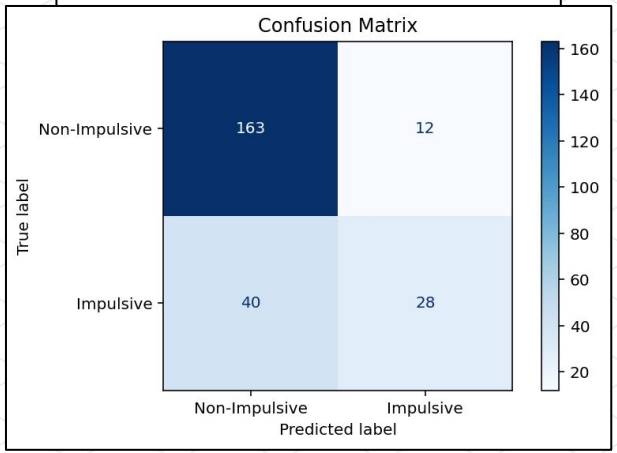
Logistic Regression (confusion matrix)

- False Negative (FN): Saying a purchase is NOT impulsive when it is - is most costly
- Backward selection has the least amount of False Negative results

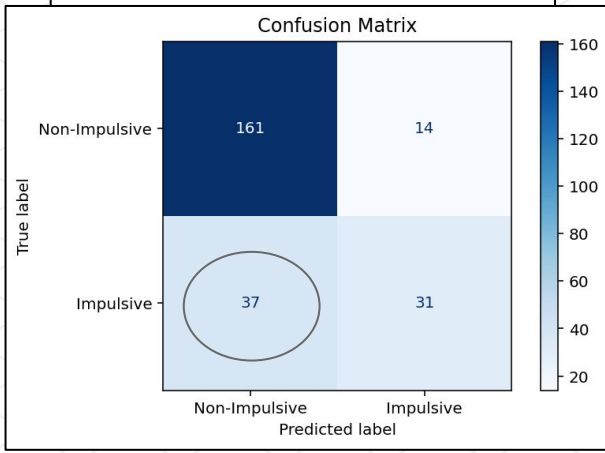
Full model



Forward selection



Backward selection



Mendeley Decision Tree

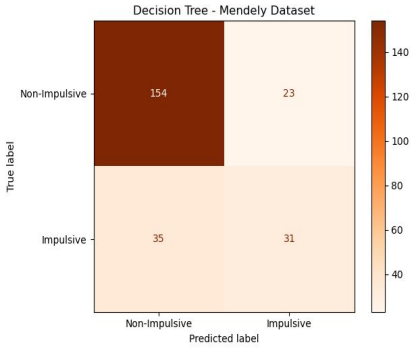
Metrics	Untuned	Tuned
Accuracy	0.7	0.76
Precision	0.46	0.57
Recall	0.55	0.47
Training F1	1	0.76
Testing F1	0.5	0.52

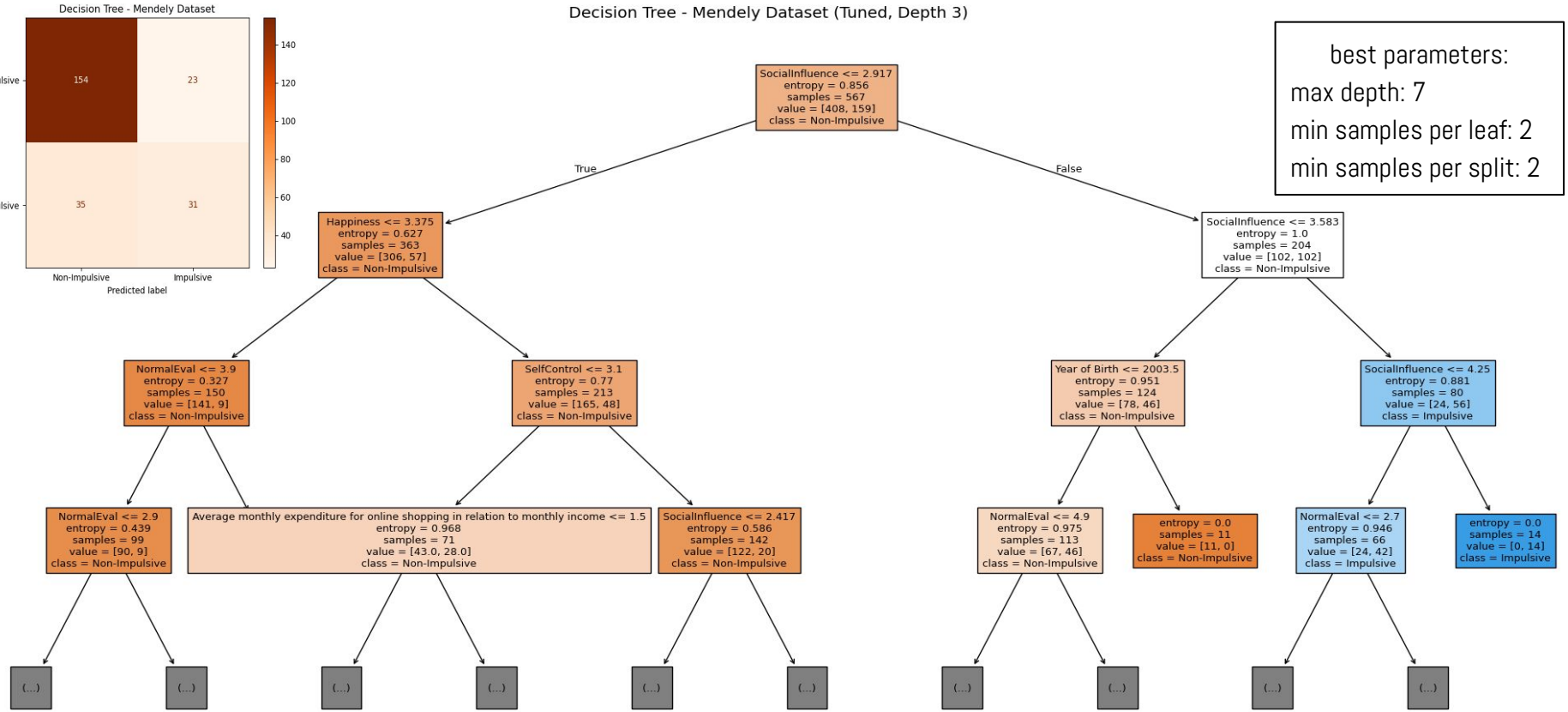
⇒ extreme overfitting, improved by tuning
⇒ tuning improved accuracy as well
⇒ low precision, recall, and f1 scores

What Did The Performance Teach Us about Impulsive Online Shoppers?

- The model identifies impulsive mindsets fairly well, but not perfectly ; **tendency, not certainty.**
- Similar precision and recall show impulsivity is not one-type; it exists on a spectrum, not binary.
- Many impulsive buyers don't self-identify as impulsive: psychology is often unconscious.

Mendeley Behavior Decision Tree





best parameters:

max depth: 7

min samples per leaf: 2

min samples per split: 2

Mendeley Random Forest

Metrics	Untuned	Tuned
Accuracy	0.79	0.78
Precision	0.65	0.63
Recall	0.46	0.47
Training F1	1	1
Testing F1	0.54	0.54

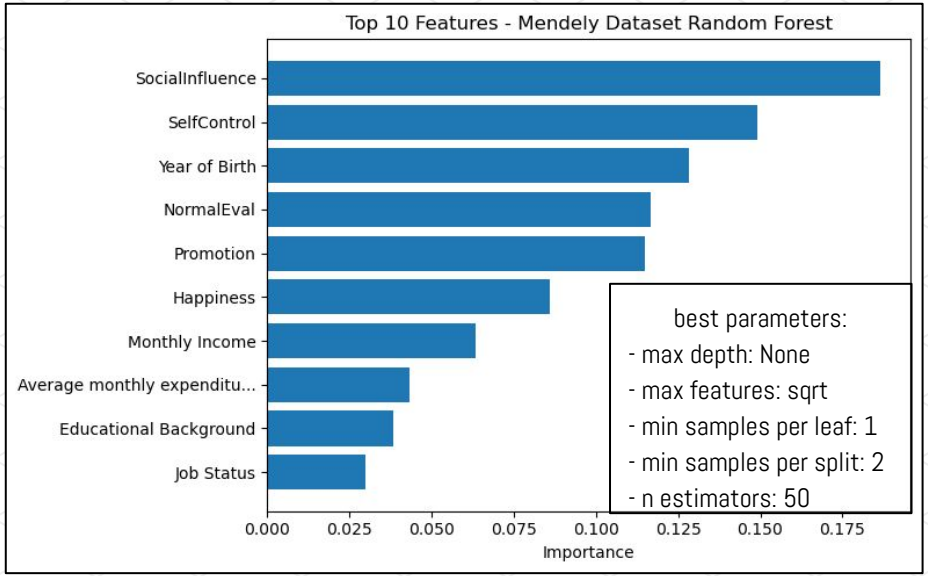
⇒ extreme overfitting on both untuned and tuned models
⇒ low recall, precision, and f1 scores
⇒ no improvement in scores between untuned and tuned

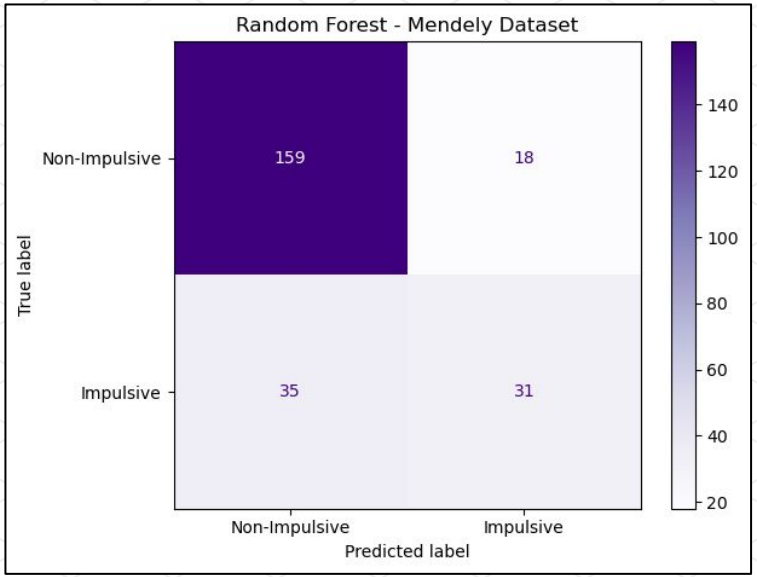
What Did The Performance Teach Us about Impulsive Online Shoppers?

- Impulsive shoppers share a strong behavioral signature detectable across many splits.
- Even with tuning, recall stayed low ⟶ impulsivity exists in more than one form.
- Suggests two shopper profiles: frequent predictable impulsives + occasional situational impulsives.

Mendeley Random Forest

Customers Driven by Social Influence & Low Self-Control are the Most Impulsive.

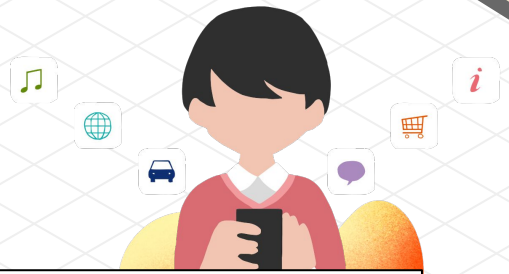




Comparing Models

Metrics	Decision Tree	Random Forest
Accuracy	0.76	0.78
Precision	0.57	0.63
Recall	0.47	0.47
Training F1	0.76	1
Testing F1	0.52	0.54

⇒ overall the models performed fairly similarly
⇒ random forest has slightly better accuracy and precision
⇒ the tuned decision tree model is better fitted than the random forest model, but both are pretty bad, with low f1 scores



Random Forest:
Many customers only show impulsive tendencies under the right emotional or contextual push.

Decision Tree:
Simpler impulsive profiles exist, but many shoppers don't fit clean rules.

KNN (scores)

Metrics	Before Tuning	After Tuning
F1 (training)	0.67	0.64
F1 (testing)	0.46	0.46
Precision	0.53	0.53
Recall	0.41	0.41
Accuracy	0.73	0.73

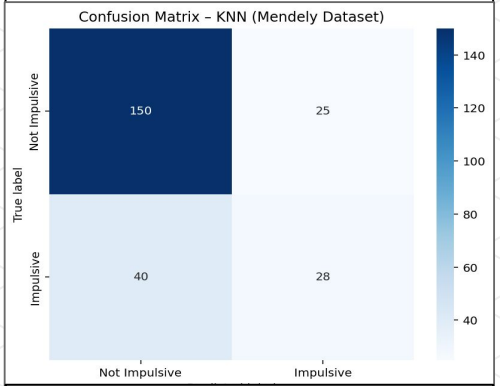
What Did The Performance Teach Us about Impulsive Online Shoppers?

Impulsivity appears in different forms:
no single "type" of impulsive customer.

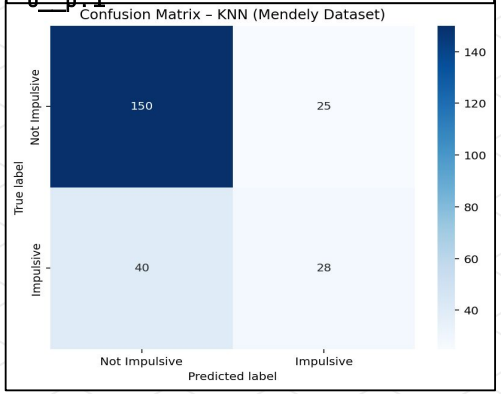
hyperparameter tuning did not provide meaningful improvements, suggesting that limitations likely came from the dataset itself rather than model settings.

KNN - Confusion Matrix

Before Tuning



After Tuning: 'b__n_neighbors': 5, 'b__p': 1



- strong at identifying **non-impulsive buyers** (150 correctly classified)
- struggles with identifying **impulsive buyers** .
  - correctly identifies 28/68 true impulsive buyers, misses 40; low recall for the impulsive class.

KNN recognizes stable psychology better than volatile psychology.

Most non-impulsive shoppers look psychologically similar, while impulsive shoppers vary too widely to fit one pattern.

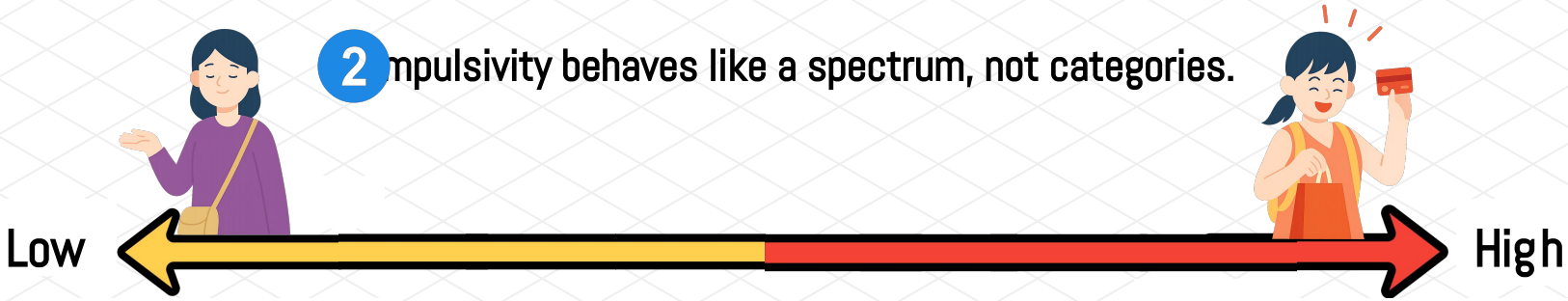
There may be multiple types of impulsive customers: some with clear impulsive traits, & others who are impulsive only in certain emotional states.

Mendeley: Conclusion

Metrics	Logistic Regression (backwards)	Decision Tree	KNN
F1 (training)	0.5747	0.755	0.64
F1 (testing)	0.5487	0.517	0.46
Precision	0.6889	0.574	0.53
Accuracy	0.7901	0.761	0.73
Recall	0.4559	0.470	0.41

1 Psychological impulsivity exists, but it is inconsistent and difficult to predict (not uniform)

2 mpulsivity behaves like a spectrum, not categories.



consistent self-control

emotional + influence-sensitive

Analysis

BEHAVIORAL	VS. PSYCHOLOGICAL
• Strongest dataset; impulsive buying is predictable from shopping behavior ⇒ leaves clear signals that allow companies to reliably flag impulsive buyers **Behavioral data predicts impulsive buying strongly, because actions like purchase frequency, promo use & subscription history give visible proof of impulse-prone patterns.**	• Model performance performed poorly ○ **Limitation: selected a higher threshold (3.5) to determine "impulsive"** ⇒ threshold can be changed according to different business models **Psychological data predicts impulsivity far less consistently; people may not act the way their traits suggest. Impulse buying is emotional, situational, and not always self-reported honestly.**

THE BOTTOM LINE:

Behavior tells who to target. Psychology tells how to influence them.

Conclusion (Business Perspective)

1

Optimize Promotions to Target Likely Impulse Shoppers using Behavioral

⇒ past purchases, promotion sensitivity, and subscription behavior reveal who is most likely to buy impulsively

- personalize promotion (flash sales, limited-time promos), predict conversion rates, increase revenue

2

Optimizing Segmentation of Likely Impulse Shoppers using Psychological

⇒ Social influence, promotion sensitivity, and lower self-control explain why buyers react, even if harder to classify

- determine psychological triggers: emotional pull → high impulsivity personas, rational framing → low impulsivity personas

Key Considerations:

- Identify the level of importance for each variable
- Avoid overly aggressive/mild tactics (ethical consideration), vice versa

Thank you!